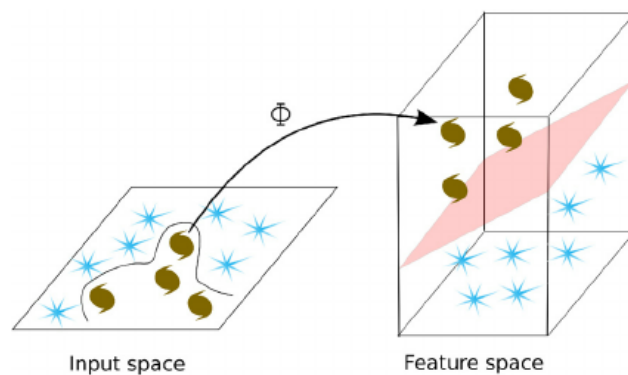

A Comparative Study of Classical and Quantum Kernel Support Vector Machines

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QUANTUM COMPUTING FOR ENGINEERS
REPORT

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1 Introduction

Quantum computing aims to introduce new computational paradigms by exploiting inherently quantum-mechanical phenomena such as superposition, entanglement and quantum interference. In recent years, these principles have attracted increasing attention in the context of machine learning, leading to the emergence of the field known as Quantum Machine Learning (QML).[1]

One of the most practically relevant and conceptually transparent approaches within QML is the use of *quantum kernel methods*. This approach allows quantum models to be integrated into well-established classical learning algorithms while modifying only the data representation [5]. Instead of replacing classical learning algorithms entirely, this approach preserves the classical optimization procedure while modifying the representation of the data through a quantum feature map. As a result, the kernel function, which measures similarity between data points, is no longer explicitly defined, but implicitly induced by the overlap of quantum states generated by a parametrized quantum circuit.

As illustrated in Figure 1, classical data is first encoded into a quantum state using a feature map implemented as a quantum circuit. The similarity between two data points is then computed as the overlap between the corresponding quantum states, which defines the quantum kernel. This kernel is subsequently used within a classical Support Vector Machine (SVM) to perform classification.

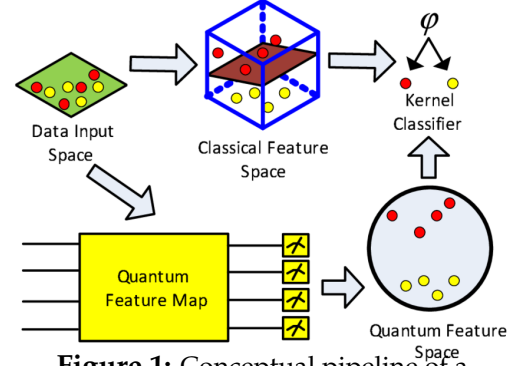


Figure 1: Conceptual pipeline of a quantum kernel method. Classical data is encoded into quantum states via a quantum feature map. The similarity between data points is computed as the overlap of quantum states, defining a quantum kernel that is subsequently used within a classical Support Vector Machine (SVM).

Support Vector Machines provide a natural framework for studying quantum kernels, since their performance is strongly determined by the choice of kernel. While classical kernels such as the linear and Radial Basis Function (RBF) kernels are well understood and highly optimized, quantum kernels offer a fundamentally different way of embedding data into high-dimensional Hilbert spaces.

The objective of this project is to perform a systematic comparison between classical and quantum kernel-based SVMs on a non-linearly separable classification task. The analysis focuses not only on predictive performance, but also on the structure of the learned decision boundaries, the expressiveness of the feature maps, and the practical limitations arising from quantum circuit simulation.

2 Classical Support Vector Machines

Support Vector Machines (SVMs) are margin-based classifiers that aim to find a separating hyperplane with maximum margin between classes. Given a labeled dataset $\{(x_i, y_i)\}_{i=1}^N$, with $y_i \in \{-1, +1\}$, the SVM seeks a decision function of the form

$$f(x) = \text{sign}(w^\top \phi(x) + b),$$

where $\phi(x)$ denotes a (possibly implicit) feature mapping and b is a bias term.

In the dual formulation, the learning problem depends only on inner products between feature vectors. This allows the introduction of kernel functions of the form

$$K(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle,$$

which implicitly define the feature space in which the classification is performed. This so-called *kernel trick* enables SVMs to learn non-linear decision boundaries without explicitly computing the feature map $\phi(x)$. A comprehensive theoretical treatment of Support Vector Machines and kernel methods can be found in [4].

Classical kernel methods are well studied and provide strong baselines for a wide range of machine learning tasks. In this project, two commonly used classical kernels are considered: the linear kernel and the Radial Basis Function (RBF) kernel.

2.1 Linear Kernel

The linear kernel is defined as

$$K(x, z) = x^\top z.$$

It does not perform any non-linear feature transformation and therefore corresponds to training a linear classifier directly in the input space. As a consequence, the learned decision boundary is a single hyperplane.

Linear SVMs are computationally efficient and easy to interpret. However, their expressive power is limited. In classification problems where class separation depends on feature interactions or higher-order correlations, linear decision boundaries are fundamentally insufficient.

Figure 2 provides an illustrative example of the geometric limitation of a linear SVM on a non-linearly separable dataset. The linear decision boundary cannot capture the underlying structure of the data, leading to systematic misclassification of samples belonging to different classes.

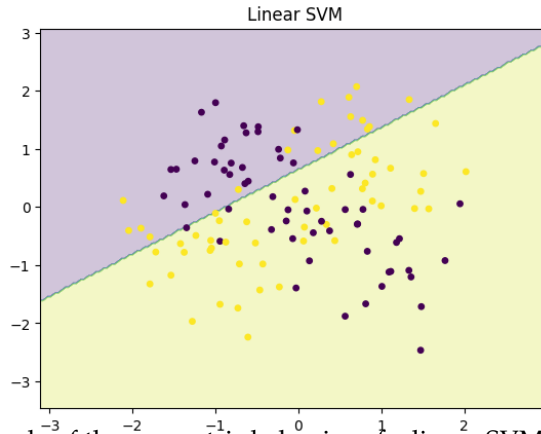


Figure 2: Illustrative example of the geometric behavior of a linear SVM on a non-linearly separable dataset. The single hyperplane is unable to correctly separate the classes.

2.2 Radial Basis Function Kernel

The Radial Basis Function (RBF) kernel is defined as

$$K(x, z) = \exp(-\gamma \|x - z\|^2),$$

where the hyperparameter γ controls the width of the Gaussian function and thus the locality of the similarity measure.

Unlike the linear kernel, the RBF kernel implicitly maps the input data into an infinite-dimensional feature space. This allows the SVM to model highly complex and non-linear decision boundaries by measuring similarity in terms of distance rather than direction.

The RBF kernel is a strong classical baseline and is widely used in practice due to its flexibility and robustness. It is particularly effective on low-dimensional problems with non-linear structure, including XOR-like classification tasks.

Figure 3 illustrates the geometric effect of the RBF kernel. The resulting decision boundary adapts smoothly to the structure of the data, demonstrating the increased expressive power of non-linear kernel methods compared to linear models.

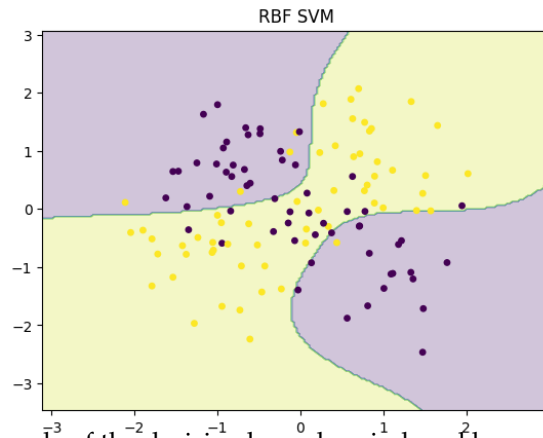


Figure 3: Illustrative example of the decision boundary induced by an SVM with RBF kernel. The non-linear kernel enables flexible separation of complex data structures.

3 Quantum Kernel Support Vector Machine

In the quantum kernel approach, classical data points are embedded into quantum states through a parametrized quantum circuit known as a *quantum feature map*. Instead of explicitly defining a feature map as in classical kernel methods, the embedding is performed implicitly by the quantum circuit itself.

Given an input vector $x \in \mathbb{R}^d$, the feature map prepares a quantum state

$$|\phi(x)\rangle = U_\phi(x) |0\rangle^{\otimes n},$$

where $U_\phi(x)$ is a data-dependent unitary operator acting on n qubits, following the standard gate-based model of quantum computation [3]. The resulting quantum state lives in a 2^n -dimensional Hilbert space, whose dimension grows exponentially with the number of qubits.

The quantum kernel is defined as the squared overlap between two such quantum states:

$$K_Q(x, z) = |\langle \phi(x) | \phi(z) \rangle|^2.$$

This definition follows the general framework of quantum kernels introduced in [5].

This quantity can be interpreted as the probability of measuring the same quantum state after applying the inverse feature map, and it naturally defines a similarity measure between data points.

Importantly, quantum kernels are guaranteed to be positive semi-definite by construction, which ensures that they can be used directly within the classical SVM optimization framework without modifying the learning algorithm.

3.1 Quantum Feature Map Design

In this project, a two-qubit feature map inspired by the ZZFeatureMap is used, originally proposed for supervised learning with quantum-enhanced feature spaces in [2]. Prior to encoding, the input features are scaled to a bounded interval in order to match the periodic nature of quantum rotation gates.

The circuit applies the following sequence of operations:

- Hadamard gates to create superposition over the computational basis,
- Single-qubit R_Z rotations to encode individual input features as phases,
- Entangling gates that introduce a phase proportional to the product $x_1 x_2$.

The inclusion of this interaction term is crucial, as it allows the quantum feature space to explicitly represent correlations between input features. This property is particularly important for XOR-type classification problems, where class membership cannot be determined from individual features alone but instead depends on their joint behavior.

In addition to the base two-qubit configuration, the feature map is extended to larger quantum systems by increasing the number of qubits. Since the input data consists of two classical features, the same encoding structure is repeated across multiple qubits. In this way, the circuits with four and eight qubits apply the same sequence of Hadamard gates, feature-dependent R_Z rotations, and entangling operations, while replicating the encoding pattern over additional qubit pairs. This strategy allows the quantum feature space to grow with the system size while preserving the same input representation.

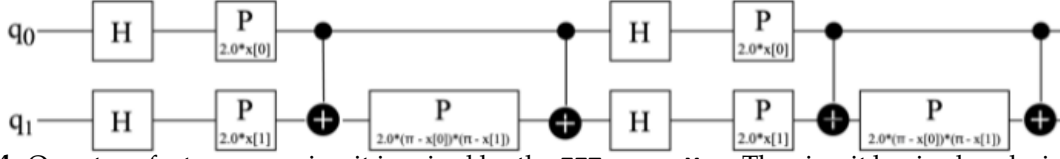


Figure 4: Quantum feature map circuit inspired by the ZZFeatureMap. The circuit begins by placing each qubit in superposition using Hadamard gates, followed by data-dependent R_Z rotations that encode the classical input features. Entangling operations introduce interaction terms between features, enabling the representation of non-linear correlations in a high-dimensional quantum Hilbert space.

3.2 Kernel Computation

Once the quantum feature map is defined, the kernel matrix is constructed by computing the pairwise overlaps between quantum states corresponding to different data points. For a training set $\{x_i\}_{i=1}^N$, this results in a kernel matrix $K \in \mathbb{R}^{N \times N}$ with entries

$$K_{ij} = K_Q(x_i, x_j).$$

Since current quantum hardware is noisy and limited in size, all quantum circuits in this project are simulated using a statevector simulator. This approach allows exact computation of quantum state overlaps and ensures that the observed behavior of the model reflects the representational power of the quantum feature map rather than hardware-induced noise.

While statevector simulation does not provide a quantum speedup, it enables controlled experimentation and systematic comparison with classical kernel methods. The resulting kernel matrix is subsequently used to train a classical Support Vector Machine with a precomputed kernel.

4 Idea and Data Generation

To meaningfully evaluate and compare kernel-based learning methods, the dataset must exhibit a non-linear decision structure that cannot be captured by simple linear models. For this reason, a synthetic XOR-type dataset is employed in this work. XOR-type datasets are commonly used to evaluate the expressive power of non-linear feature mappings in both classical and quantum learning settings [5].

Data points are sampled uniformly in a two-dimensional input space, $x = (x_1, x_2) \in \mathbb{R}^2$, and class labels are assigned according to the sign of the product of the two features:

$$y = \text{sign}(x_1 x_2).$$

As a result, samples belong to one class when the features have the same sign and to the opposite class when their signs differ. This labeling rule produces the well-known XOR structure, which is not linearly separable and explicitly depends on feature interactions rather than individual feature values.

To increase the difficulty of the task and avoid trivial separation, additive Gaussian noise is introduced in the feature space. This creates overlapping regions between classes and makes the classification problem more realistic and challenging.

Figure 5 illustrates the resulting dataset after standardization. The presence of overlapping clusters highlights the limitations of linear classifiers and motivates the use of non-linear kernel methods. In particular, this setting is well suited for investigating the expressive power of both classical and quantum kernels.

The dataset is split into training and test sets using stratified sampling in order to preserve class balance across both subsets.

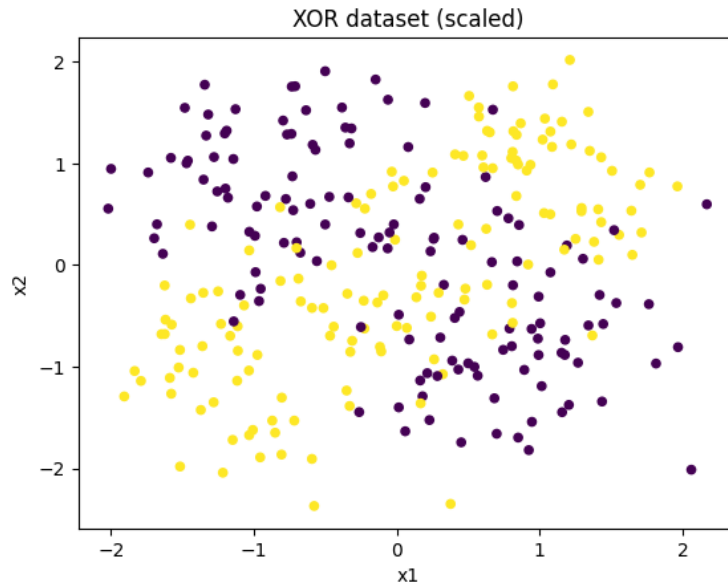


Figure 5: Standardized XOR dataset used in this work. Data points are colored according to their class labels. The non-linear structure and overlapping regions between classes make the problem unsuitable for linear classification and motivate the use of kernel-based methods.

5 Results and Discussion

This section presents a detailed comparison between several classification models: a linear Support Vector Machine (SVM), a classical SVM with Radial Basis Function (RBF) kernel, and Quantum Kernel SVMs implemented with different numbers of qubits (2, 4, and 8). The objective is to analyze both quantitative performance metrics and qualitative properties of the learned decision functions.

Model performance is evaluated on an unseen test set using accuracy and F1-score, which together provide a balanced assessment of classification quality, particularly in the presence of class overlap.

5.1 Quantitative Results

Table 1 summarizes the quantitative performance of the evaluated models, including the classical baselines and the Quantum Kernel SVMs implemented with different qubit counts. As expected, the linear SVM achieves the lowest accuracy and F1-score. This behavior is a direct consequence of the linear nature of its decision boundary, which is fundamentally incapable of representing the XOR structure of the dataset. The classical RBF SVM achieves the best overall performance. Its

Table 1: Comparison of classification performance on the test set.

Model	Accuracy	F1-score
Linear SVM	0.575	0.633
RBF SVM	0.825	0.842
Quantum Kernel SVM (2 qubits)	0.692	0.726
Quantum Kernel SVM (4 qubits)	0.742	0.744
Quantum Kernel SVM (8 qubits)	0.750	0.758

high accuracy and F1-score demonstrate the expressive power of classical kernel methods when applied to low-dimensional, non-linearly separable problems. This result is consistent with the well-known effectiveness of RBF kernels on low-dimensional non-linear problems [4].

The Quantum Kernel SVM exhibits intermediate performance. While the two-qubit model already improves over the linear SVM, increasing the number of qubits to four and eight leads to a moderate improvement in performance. This behavior suggests that larger quantum feature spaces can provide additional representational capacity. However, even with increased qubit counts, the quantum kernel does not surpass the classical RBF kernel in this experimental setting.

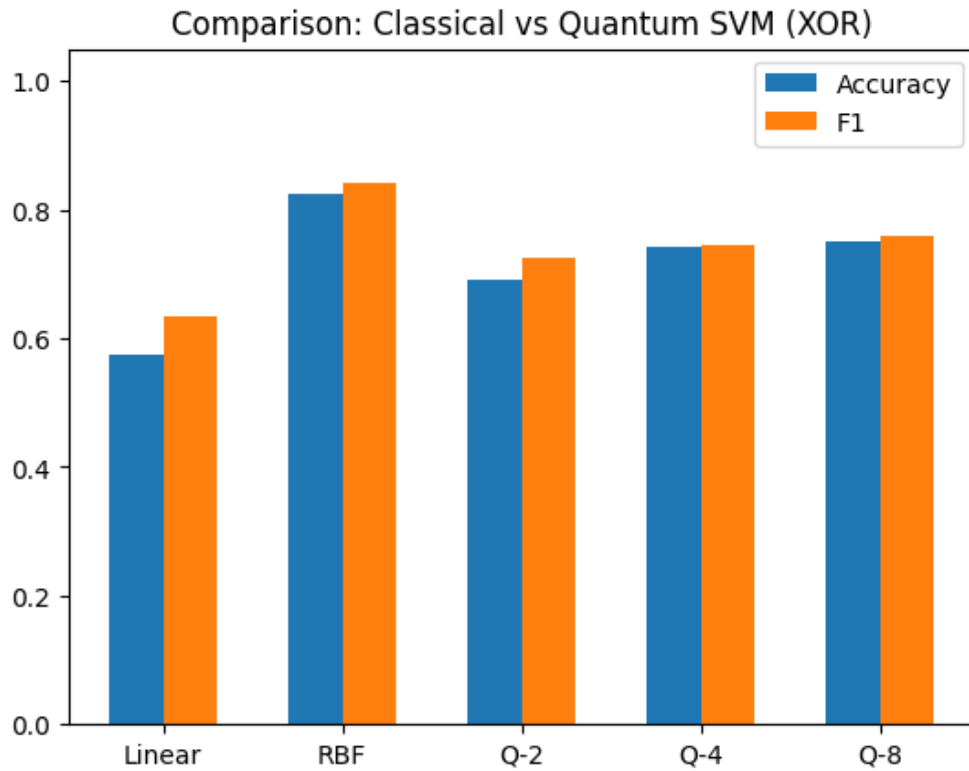


Figure 6: Comparison of Accuracy and F1-score for classical SVM models and Quantum Kernel SVMs with different qubit counts (2, 4, and 8). The RBF SVM achieves the highest performance, while the quantum models show intermediate performance that improves as the number of qubits increases.

5.2 Confusion Matrix Analysis

Confusion matrices provide insight into the types of errors made by each model and complement aggregate performance metrics such as accuracy and F1-score.

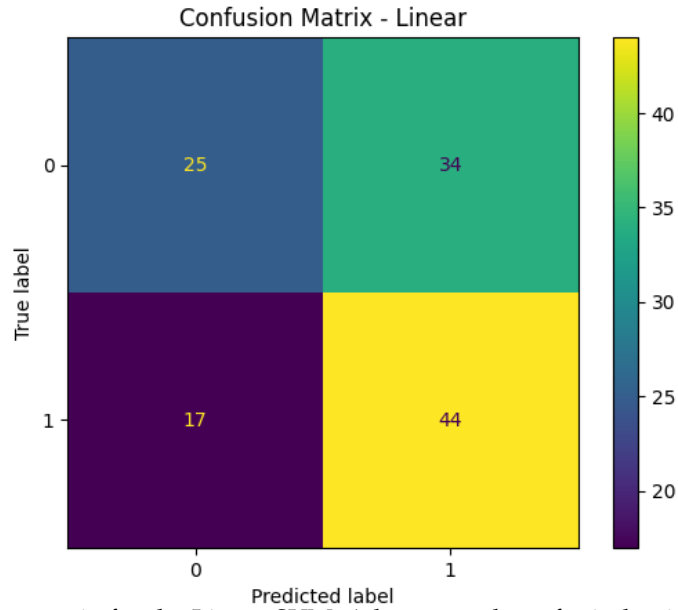


Figure 7: Confusion matrix for the Linear SVM. A large number of misclassifications is observed, reflecting the inability of linear decision boundaries to represent the XOR structure of the data.

Overall, the confusion matrices confirm the trends observed in the quantitative metrics. The Linear SVM produces a large number of misclassifications due to its inability to model non-linear decision boundaries. The RBF SVM achieves the most accurate classification, correctly identifying the majority of samples in both classes. The Quantum Kernel SVM reduces the number of errors compared to the linear baseline, and the results show a gradual improvement when increasing the number of qubits from two to four and eight. This behavior suggests that larger quantum feature spaces can enhance the representational capacity of the model, although the classical RBF kernel still achieves the best performance in this experiment.

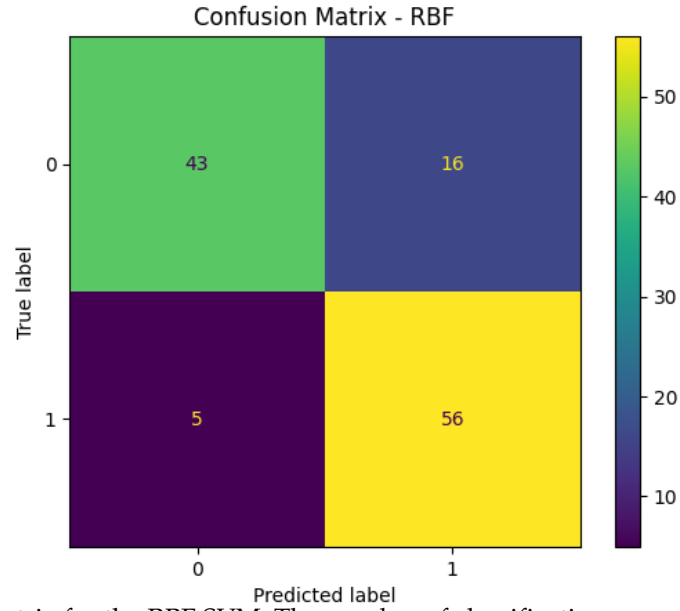


Figure 8: Confusion matrix for the RBF SVM. The number of classification errors is significantly reduced, demonstrating the effectiveness of the RBF kernel in separating non-linearly structured data.

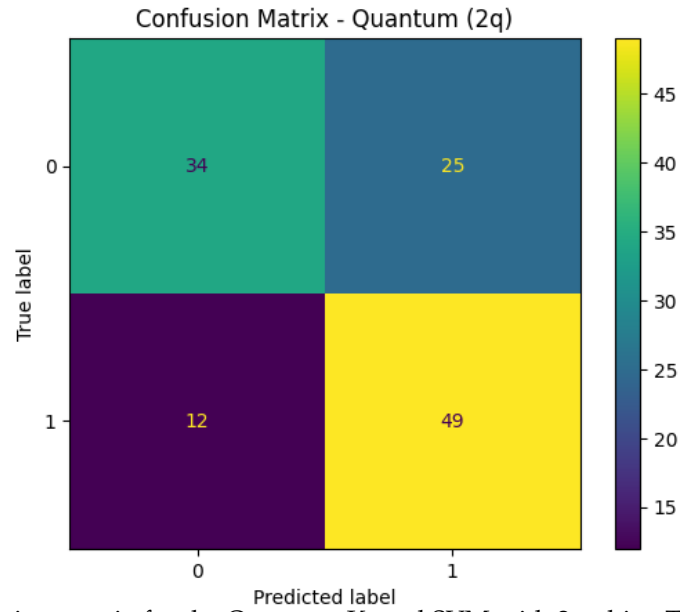


Figure 9: Confusion matrix for the Quantum Kernel SVM with 2 qubits. The model reduces misclassification compared to the linear baseline, indicating that the quantum feature map captures non-linear feature interactions, although performance remains below the classical RBF kernel.

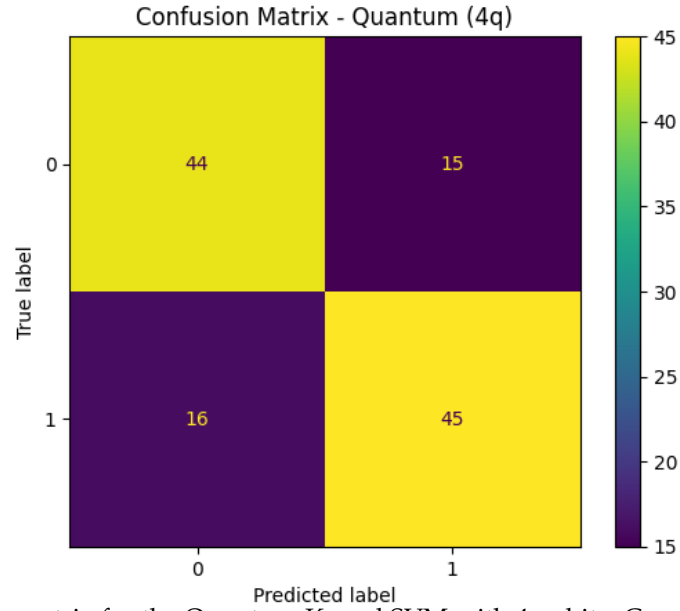


Figure 10: Confusion matrix for the Quantum Kernel SVM with 4 qubits. Compared to the two-qubit model, the number of correctly classified samples increases slightly, indicating that the larger quantum feature space improves the representation of the data structure.

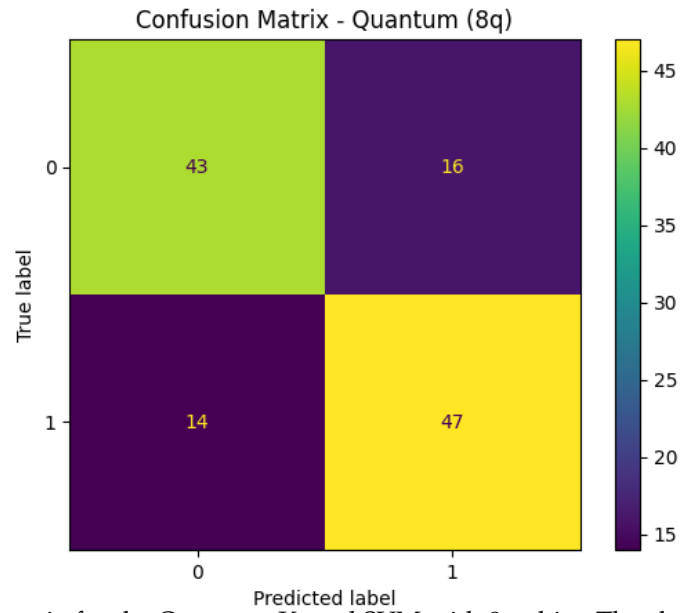


Figure 11: Confusion matrix for the Quantum Kernel SVM with 8 qubits. The classification results further improve compared to the smaller quantum models, although the performance still remains below that of the classical RBF kernel.

5.3 Decision Boundary Analysis

Visual inspection of the learned decision boundaries provides qualitative insight into how each model separates the input space.

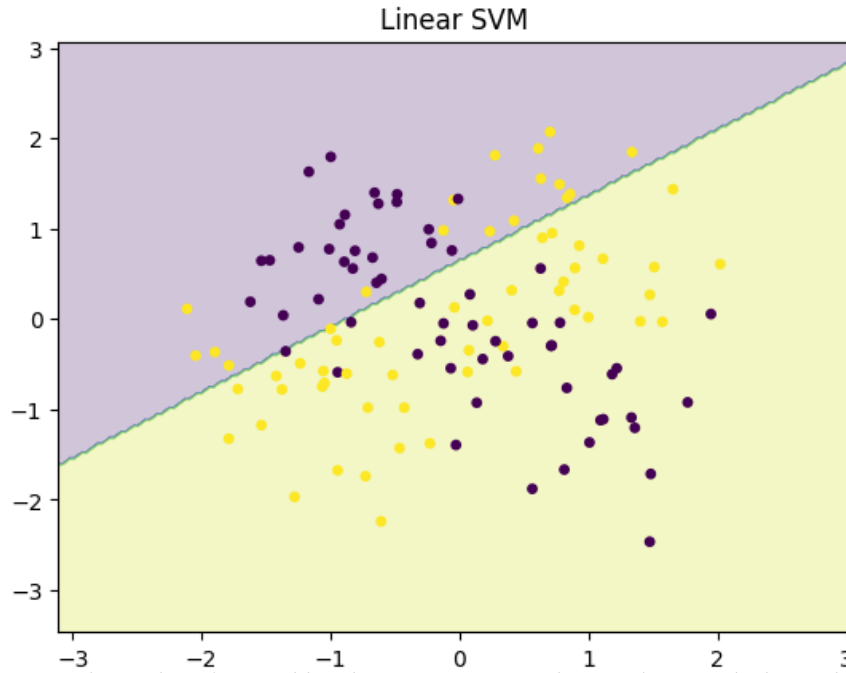


Figure 12: Decision boundary learned by the Linear SVM. The single straight boundary is unable to capture the XOR structure of the dataset.

Overall, the decision boundary visualizations are consistent with the quantitative results. The linear model fails to capture the XOR structure due to its limited expressiveness, while the RBF kernel produces a smooth boundary closely aligned with the underlying class distribution. The quantum models generate increasingly complex decision regions as the number of qubits increases, indicating that larger quantum feature spaces allow richer representations of non-linear feature interactions.

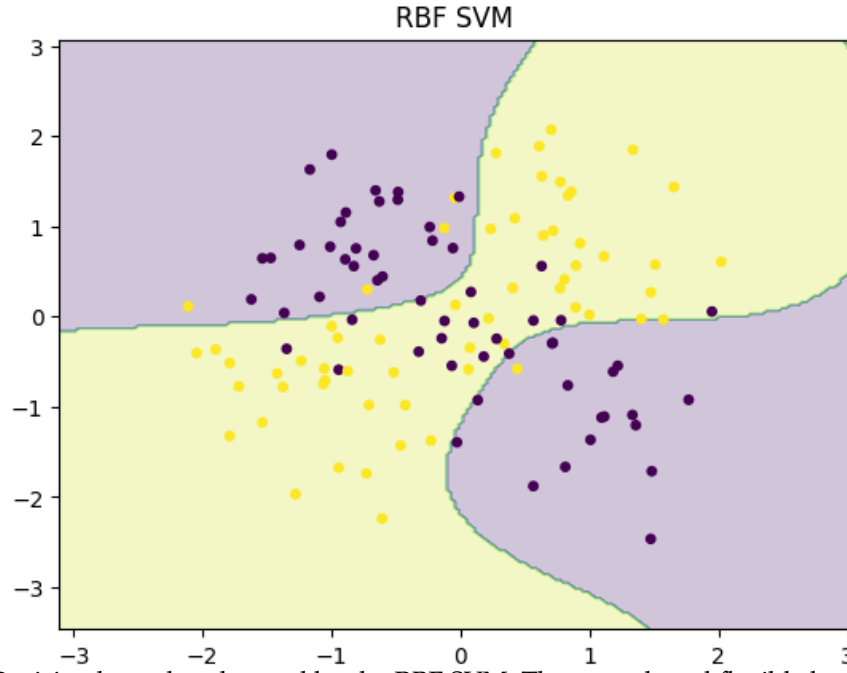


Figure 13: Decision boundary learned by the RBF SVM. The smooth and flexible boundary closely follows the true class structure of the XOR dataset.

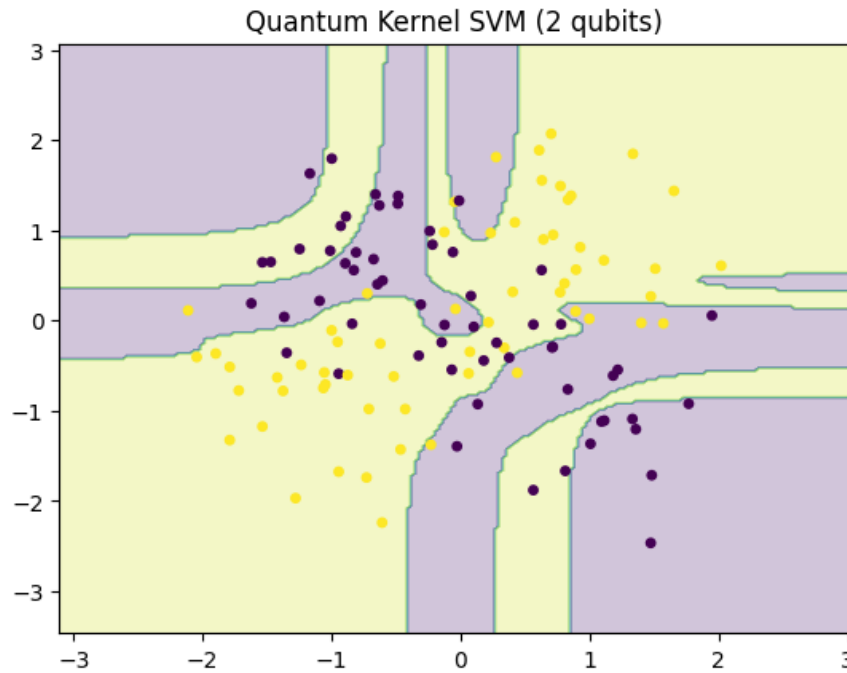


Figure 14: Decision boundary learned by the Quantum Kernel SVM with 2 qubits. The resulting regions reflect the non-linear feature interactions encoded by the quantum feature map, improving over the linear baseline.

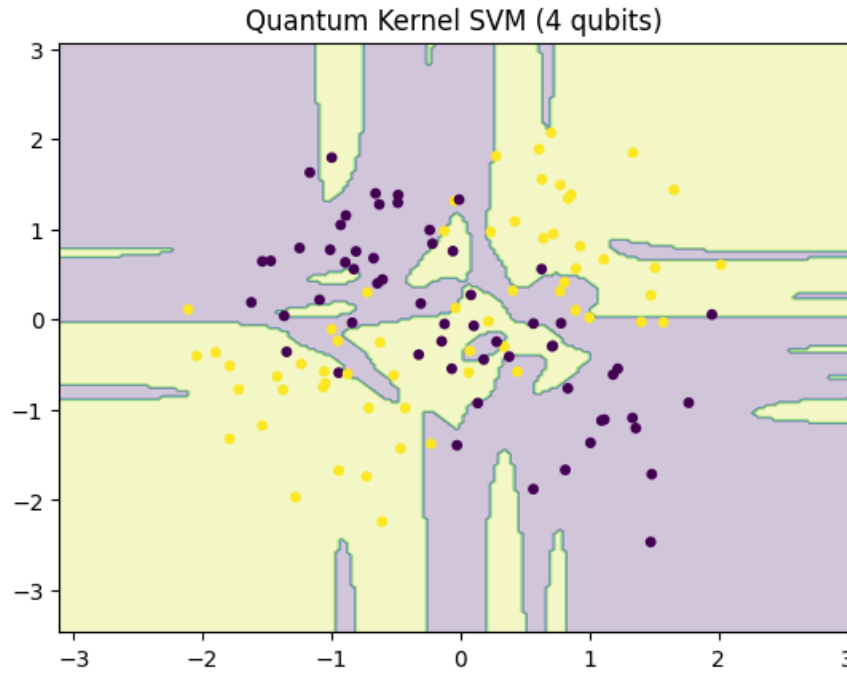


Figure 15: Decision boundary learned by the Quantum Kernel SVM with 4 qubits. Increasing the number of qubits produces a richer partitioning of the input space, leading to improved alignment with the XOR structure.

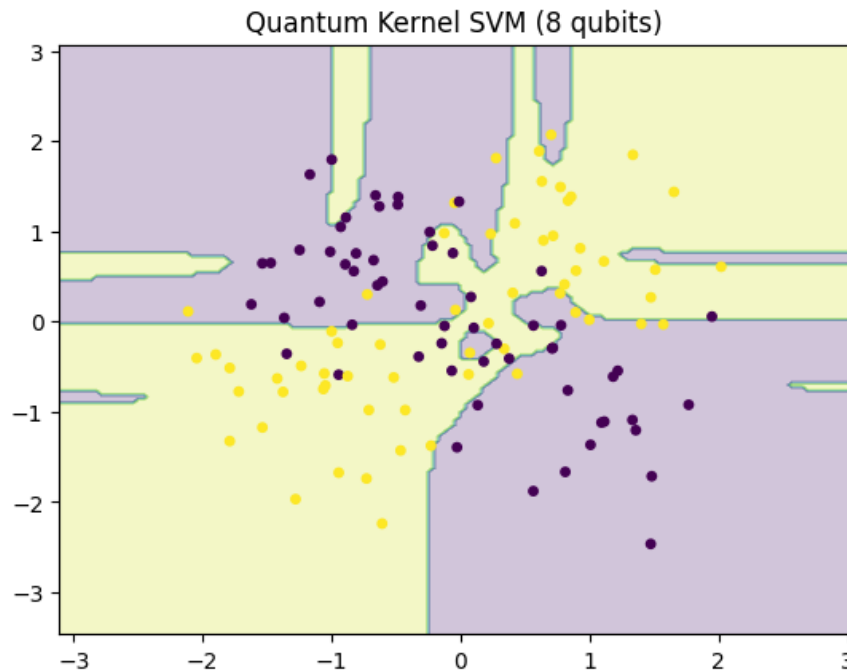


Figure 16: Decision boundary learned by the Quantum Kernel SVM with 8 qubits. The higher-dimensional quantum feature space allows more complex decision regions, although the resulting boundary remains less smooth than the classical RBF model.

6 Interpretation and Limitations

The experimental results do not demonstrate a clear quantum advantage over classical kernel methods. This observation is in line with previous experimental studies on quantum kernels, which report expressive feature maps but no systematic advantage on near-term quantum devices [2]. In particular, while the Quantum Kernel SVM consistently outperforms the linear SVM, it does not surpass the classical RBF kernel in terms of predictive performance. This outcome, however, is not unexpected and should be interpreted with care.

The primary objective of this project is not to establish superiority of quantum methods over classical approaches, but rather to analyze how quantum kernels behave in practice and to identify the conditions under which they may become competitive. From this perspective, the results provide valuable insight into both the strengths and current limitations of quantum kernel-based learning.

A key observation is that the quantum kernel is able to capture non-linear feature interactions that are inaccessible to linear models. This is particularly evident in the XOR classification task, where the class labels depend explicitly on the joint behavior of the input features. The use of entangling gates in the quantum feature map enables the representation of such interactions in a natural and structured way, leading to a clear improvement over linear decision boundaries.

In order to further investigate the representational capacity of the quantum feature space, the feature map was also implemented with larger quantum systems consisting of four and eight qubits. Since the input data contains only two classical features, the same encoding structure was repeated across multiple qubit pairs. The experimental results show a gradual improvement in classification performance when increasing the number of qubits, indicating that larger quantum feature spaces can provide additional representational capacity. However, the improvement remains moderate and does not surpass the performance of the classical RBF kernel.

At the same time, several important limitations constrain the performance of the quantum model in this setting:

- **Limited number of qubits.** The dimensionality of the quantum feature space grows exponentially with the number of qubits. In this work, the feature map was implemented with two, four, and eight qubits in order to analyze how increasing the system size affects the resulting kernel. While larger quantum systems provide a richer feature space and lead to modest improvements in classification performance, the explored system sizes remain relatively small and therefore cannot match the flexibility of highly expressive classical kernels.
- **Circuit depth and expressiveness.** The structure and depth of the quantum circuit play a crucial role in determining the quality of the induced kernel. Shallow circuits are chosen to ensure numerical stability and tractability in simulation, but this also restricts the complexity of the feature map. Designing deeper or more expressive circuits may improve performance, but introduces additional challenges related to noise and optimization.
- **Strength of classical baselines.** Classical kernels such as the RBF kernel operate in implicitly infinite-dimensional feature spaces and have been refined and optimized over decades of research. As a result, they provide extremely strong baselines, particularly for low-dimensional problems such as the one considered in this project. Surpassing such methods with near-term quantum models remains a significant challenge.

In addition to these conceptual limitations, practical constraints also play an important role. All quantum computations in this project are performed using a statevector simulator, which eliminates hardware noise but also removes any potential computational advantage. Consequently,

the results should be interpreted as an analysis of representational power rather than execution speed or scalability.

Despite these limitations, the findings confirm that quantum kernels can serve as meaningful and well-defined feature representations within classical learning algorithms. The observed improvement over linear models demonstrates that even small quantum circuits can encode useful non-linear structure. As quantum hardware continues to improve and allows for larger, deeper, and less noisy circuits, quantum kernel methods may become increasingly competitive in more complex learning tasks.

Overall, this work highlights both the promise and the current boundaries of quantum kernel-based machine learning, and provides a clear foundation for future research exploring more expressive feature maps, larger quantum systems, and hybrid classical– quantum learning strategies. Quantum kernel methods were first systematically introduced in [2], where quantum feature maps were shown to provide expressive kernels for classification tasks.

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